

- $\ddot{u} + u - \frac{1}{a} - \delta u^2 = 0$

- $u(\theta) = \frac{1}{r(\theta)}$

$$\boxed{1} \quad \ddot{u} = \delta u^2 - u + \frac{1}{a}$$

$$\Rightarrow \begin{cases} \dot{u}(\theta) = r(\theta) \\ \dot{r}(\theta) = \delta u^2(\theta) - u(\theta) + \frac{1}{a} \end{cases}$$

$$2] \underline{g=0}$$

$$\therefore \begin{cases} \ddot{u}(\theta) = v(\theta) \\ \dot{v}(\theta) = -u(\theta) + \frac{1}{a} \end{cases}$$

$$\Rightarrow \ddot{u} + u - \frac{1}{a} = 0 \Rightarrow \ddot{u} + u = \frac{1}{a}$$

Buho $\text{Solución general} = \text{Solución homogénea} + \text{Solución Particular}$

Sol. homogénea:

$$\ddot{u} + u = 0$$

$$\text{Vamos } u = e^{\lambda \cdot \theta} \quad \text{tg: } \begin{cases} \dot{u} = \lambda \cdot e^{\lambda \cdot \theta} \\ \ddot{u} = \lambda^2 \cdot e^{\lambda \cdot \theta} \end{cases}$$

$$\therefore \lambda^2 \cdot e^{\lambda \cdot \theta} + e^{\lambda \cdot \theta} = 0 \Rightarrow$$

$$\Rightarrow \lambda^2 + 1 = 0 \Rightarrow \lambda^2 = -1$$

$$\Rightarrow \lambda = \pm \sqrt{-1} = \pm i$$

Per la forma:

$$V_h(\theta) = e^{i \cdot \theta} \left[A \cdot \cos(\theta) + B \cdot \sin(\theta) \right]$$

$$\therefore V_h(\theta) = A \cdot \cos(\theta) + B \cdot \sin(\theta)$$

Sol. puntual:

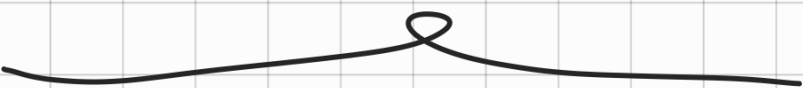
Ahora; como $V_p = C$, con C una constante y $V_H + V_p = \frac{1}{a}$:

$$\Rightarrow V_H + V_p = 0 + C = \frac{1}{a} \Rightarrow C = \frac{1}{a}$$

$$\Rightarrow V_p = \frac{1}{a}$$

Par le théorème,

$$V(\theta) = A \cdot \cos(\theta) + B \cdot \sin(\theta) + \frac{1}{a}$$



Ainsi bien,

$$V(\theta) = \frac{a}{1+\varepsilon} \quad ; \quad V(\theta) = \frac{1}{V(\theta)}$$

$$\Rightarrow \frac{a}{1+\varepsilon} = \frac{1}{A \cdot \cos(\theta) + B \sin(\theta) + \frac{1}{a}}$$

$$\frac{a}{1+\varepsilon} = \frac{1}{A + \frac{1}{a}}$$

$$a \cdot A + 1 = 1 + \varepsilon$$

$$a \cdot A = \varepsilon$$

$$A = \frac{\varepsilon}{a}$$

Luego,

$$\dot{r}(0) = 0$$

$$\therefore \dot{r} = \frac{dr}{d\theta} \quad \text{y como } r \text{ es una función}$$

conjugada:

$$\therefore \dot{r} = \frac{dr}{d\theta} = \frac{dr}{du} \cdot \frac{du}{d\theta}$$

$$\Rightarrow \dot{r}(0) = -\frac{1}{u(0)} \cdot \frac{du}{d\theta}(0)$$

$$\therefore \text{como } -\frac{1}{u(0)} \neq 0 \Rightarrow \frac{du}{d\theta}(0) = 0 = \dot{u}(0)$$

$$\therefore \dot{u}(\theta) = -A \cdot \operatorname{sen}(\theta) + B \cdot \operatorname{cos}(\theta)$$

$$\Rightarrow \dot{u}(0) = -A \cdot \operatorname{sen}(0) + B \cdot \operatorname{cos}(0) = 0$$

$$0 + B = 0 \Rightarrow B = 0$$

Por lo tanto,

$$V(\theta) = \frac{\varepsilon}{a} \cdot \cos(\theta) + \frac{1}{a} = \frac{1}{a} \cdot (\varepsilon \cdot \cos(\theta) + 1)$$

$$\Rightarrow r(\theta) = \frac{a}{\varepsilon \cdot \cos(\theta) + 1}$$

Siguiendo las condiciones iniciales, $a=1$:

$$\therefore r(\theta) = \frac{1}{\varepsilon \cdot \cos(\theta) + 1}$$

Hay un situación límite donde el denominador
tiende a cero:

$$\varepsilon \cdot \cos(\theta) + 1 = 0 \Rightarrow \varepsilon \cdot \cos(\theta) = -1$$

$$\Rightarrow \cos(\theta) = -\frac{1}{\varepsilon} \Rightarrow \theta_x = \arccos\left(-\frac{1}{\varepsilon}\right)$$

$$\therefore \left| -\frac{1}{\varepsilon} \right| = \frac{1}{\varepsilon} \leq 1 \Rightarrow \varepsilon \geq 1$$

"∴ El denominador puede hacerse 0 cuando $\varepsilon \neq 1$.

$$3] \begin{cases} \ddot{U}(\theta) = r(\theta) \\ \dot{r}(\theta) = \int U^2(\theta) - U(\theta) + \frac{1}{a} \end{cases}$$

$$\Rightarrow \ddot{U} + U = \frac{1}{a} + \int U^2$$

Veamos que $\ddot{U} + U = \frac{1}{a} \Rightarrow U_0(\theta) = \frac{1}{a} (\varepsilon \cos(\theta) + 1)$

$$\Rightarrow U(\theta) = U_0(\theta) + \int U^2$$

$\therefore$ Los regímenes de ε son los mismos.